

49 Amplified Generalized Fractional Neural Network Approximations at a reduced finite domain

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Abstract: This research deals with the determination of the rate of generalized fractional pointwise, uniform and L_p , $p \geq 1$, convergences to the unit operator of 49 specific "normalized cusp neural network operators". The cusp is a compact support activation function, which derives from the composition of two specific activation functions having as domain the whole real line. We employ 7 different known activation functions. These convergences are given via the moduli of continuity of the right and left generalized fractional derivatives of Caputo type of the engaged function in the form of Jackson type inequalities. The composition of activation functions aims to more flexible and powerful neural networks utilizing the reduction of infinite domains to the one domain of compact support.

Keywords: Amplified Neural network approximation, cusp activation function, modulus of continuity, reduction of domain, generalized fractional derivatives.

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1 Introduction

From AI and computer science we have the following: In essence, composing activation functions in neural networks offers the advantage of potentially tailoring the network's ability to learn and model complex, non-linear relationships in data. Here's a breakdown of the potential benefits:

1. Enhanced Capacity for Complex Modeling:

–Diversification of Non-linearity: Different activation functions have different characteristics. For example, ReLU introduces sparsity, while Sigmoid squashes values into a range. By composing them, the network potentially can learn a wider variety of non-linear transformations and capture more intricate patterns in the data.

2. Improved Training Dynamics:

–Mitigating Gradient Problems: Activation functions influence gradient flow during training. Using different activation functions can potentially help address issues like vanishing or exploding gradients, which hinder learning in deep networks.

–Faster Convergence: Certain activation functions, like ReLU, can accelerate the convergence of the training process compared to others like Sigmoid or Tanh. Combining different functions can potentially lead to faster training and competitive performance.

3. Enhanced Generalization and Robustness:

–Better Generalization: By learning richer representations of the data through diverse activation functions, the network's ability to generalize well to unseen data improves, reducing the risk of overfitting.

–Increased Robustness: Networks with carefully chosen activation functions can handle variations in input data more effectively, adapting to noise, missing data, or unexpected perturbations.

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4. Adaptation to Input Characteristics:

–Handling Diverse Data: Different activation functions can be suited to different data characteristics. For instance, tanh can be useful when dealing with data containing both positive and negative values.

5. Potential for Architectural Interpretability:

–Insight into Learning: By using distinct activation functions, different parts of the network might become responsible for capturing specific features, which can potentially offer insights into how the model learns.

In summary, composing activation functions potentially allows for a more flexible and powerful neural network capable of:

- Learning more complex patterns.
- Faster and more stable training.
- Better generalization to new data.
- Greater adaptability to diverse data.

Attention: While composing activation functions can offer benefits, it's important to choose them judiciously and with consideration for the specific problem at hand, as some combinations might not be beneficial or could even lead to unwanted behaviors like exploding gradients. Empirical testing and validation are crucial when exploring different activation function compositions.

The author was greatly inspired and motivated by [8] and he was the pioneer of quantitative neural network approximation, see [2], and since then he has published numerous of papers and books, e.g. see [3]- [7].

In this article we continue this trend at the generalized fractional level.

In mathematical neural network approximation AMS Mathscinet lists no articles related to composition of activation functions. So this is a seminal article.

By using composition of known and specific activation functions he achieves the first long part of the introduction, and most notably these compositions lead to activation functions of compact support, though the initial activation functions had an infinite domain, the whole real line.

Now each resulting bi-composed activation function is an open cusp of compact support $[-1, 1]$. Our involved 7 activation functions generate 49 amplified neural network operators, which resemble to the squashing operators in [2], [4], and so do the produced quantitative results.

Of great inspiration are the articles [12]-[14]. References [15], [16], [18] are foundational. Finally references [19]-[28] represent recent important works.

2 Foundations

We need

Definition 1.([7], p. 530) *The sigmoid activation functions we deal with here are described as follows $g : \mathbb{R} \rightarrow [-1, 1]$, strictly increasing $g(0) = 0$, $g(-x) = -g(x)$, $x \in \mathbb{R}$, $g(+\infty) = 1$, $g(-\infty) = -1$. Also g is strictly convex over $(-\infty, 0]$ and strictly concave over $[0, +\infty)$, with $g^{(2)} \in C(\mathbb{R})$.*

More precisely we will deal with the following sigmoid activation functions which possess all of the above properties: the algebraic activation function,

$$g_1(x) = \frac{x}{\sqrt[m]{1+x^{2m}}}, \quad m \in \mathbb{N}, x \in \mathbb{R}, \quad (1)$$

(see [6], pp. 2-3);

the normalized and parametrized arctangent

$$g_2(x) = \frac{2}{\pi} \arctan\left(\frac{x}{2}\lambda x\right) = \frac{2}{\pi} \int_0^{\frac{\pi\lambda x}{2}} \frac{dz}{1+z^2}, \quad x \in \mathbb{R}, \lambda > 0, \quad (2)$$

(see [7], p.p. 156-157);

the normalized and parametrized Gudermannian function

$$g_3(x) = \frac{2}{\pi} gd(\lambda x) = \frac{4}{\pi} \arctan\left(\tanh\left(\frac{\lambda x}{2}\right)\right)$$

$$= \frac{2}{\pi} \int_0^{\lambda x} \frac{dt}{\cosh t} = \frac{4}{\pi} \int_0^{\lambda x} \frac{dt}{e^t + e^{-t}}, \quad x \in \mathbb{R}, \lambda > 0, \quad (3)$$

where

$$gd(x) := \int_0^x \frac{dt}{\cosh t} = 2 \arctan \left(\tanh \left(\frac{x}{2} \right) \right), \quad \forall x \in \mathbb{R},$$

is the Gudermannian function (see [7], p. 198-199);

the parametrized error function

$$g_4(x) = \operatorname{erf} \lambda x = \frac{2}{\pi} \int_0^{\lambda x} e^{-t^2} dt, \quad x \in \mathbb{R}, \lambda > 0, \quad (4)$$

(see [7], pp. 226-227);

the parametrized hyperbolic tangent function

$$\begin{aligned} g_5(x) = \tanh \lambda x &= \frac{e^{\lambda x} - e^{-\lambda x}}{e^{\lambda x} + e^{-\lambda x}} = \frac{e^{2\lambda x} - 1}{e^{2\lambda x} + 1} \\ &= \frac{1 - e^{-2\lambda x}}{1 + e^{-2\lambda x}}, \quad x > 0, \lambda > 0, \end{aligned} \quad (5)$$

(see [7], p. 72);

for small $0 < \lambda < 1$, $\tanh \lambda x$ is expected to behave better than *ReLU* and *Leaky ReLU* activation functions;

the parametrized hyperbolic tangent like activation function,

$$g_6(x) = \frac{A^{\lambda x} - A^{-\lambda x}}{A^{\lambda x} + A^{-\lambda x}}, \quad A > 1, \lambda > 0, x \in \mathbb{R} \quad (6)$$

(see [7], pp. 262-263);

finally, the β -parametrized half hyperbolic tangent function

$$g_7(x) = \frac{1 - e^{-\beta x}}{1 + e^{-\beta x}}, \quad \beta > 0, \forall x \in \mathbb{R}, \quad (7)$$

(see [7], p. 509-510).

So, we can denote then by $g_i(x)$, $x \in \mathbb{R}$, $i = 1, 2, \dots, 7$.

Here we are concerned with the compositions of these activation functions

$$g_i \circ g_j = g_i|_{(-1,1)} \circ g_j, \quad \text{and} \quad g_i \circ g_i = g_i|_{(-1,1)} \circ g_i, \quad (8)$$

where $i, j \in \{1, 2, \dots, 7\}$.

There are $P(7, 2)$ permutations, i.e.

$$P(7, 2) = \frac{7!}{5!} = 6 \cdot 7 = 42, \text{ of ordered compositions;}$$

plus 7 self-compositions; so we have a total of 49 total such compositions.

So here a $g_i \circ g_j$ means any of the compositions in (8), where $i, j \in \{1, \dots, 7\}$, $i \neq j$ or $i = j$.

Clearly, $g_i \circ g_j = g_i|_{(-1,1)} \circ g_j$ is strictly increasing and $(g_i \circ g_j)(0) = 0$, and

$$g_i \circ g_j(-x) = g_i(g_j(-x)) = g_i(-g_j(x)) = -g_i(g_j(x)) = -(g_i \circ g_j)(x),$$

that is

$$(g_i \circ g_j)(-x) = -(g_i \circ g_j)(x), \quad \forall x \in \mathbb{R}.$$

Furthermore

$$\begin{aligned} (g_i \circ g_j)(+\infty) &= g_i(g_j(+\infty)) = g_i(1), \\ (g_i \circ g_j)(-\infty) &= g_i(g_j(-\infty)) = g_i(-1). \end{aligned}$$

Next acting over $(-\infty, 0]$: let $\lambda, \mu \geq 0$: $\lambda + \mu = 1$. Then by convexity of g_j there we have

$$g_j(\lambda x + \mu y) \leq \lambda g_j(x) + \mu g_j(y), \quad x, y \in \mathbb{R}_-;$$

and

$$g_i(g_j(\lambda x + \mu y)) \leq g_i(\lambda g_j(x) + \mu g_j(y)) \leq \lambda g_i(g_j(x)) + \mu g_i(g_j(y)),$$

i.e.

$$(g_i \circ g_j)(\lambda x + \mu y) \leq \lambda (g_i \circ g_j)(x) + \mu (g_i \circ g_j)(y),$$

$x, y \in \mathbb{R}_-$.

So that $g_i \circ g_j$ is convex over $(-\infty, 0]$.

Similarly, over $[0, +\infty)$ we get: let $\lambda, \mu \geq 0 : \lambda + \mu = 1$. Then by concavity of g_j there we have

$$g_j(\lambda x + \mu y) \geq \lambda g_j(x) + \mu g_j(y), \quad x, y \in \mathbb{R}_+;$$

and

$$g_i(g_j(\lambda x + \mu y)) \geq g_i(\lambda g_j(x) + \mu g_j(y)) \geq \lambda g_i(g_j(x)) + \mu g_i(g_j(y)).$$

Therefore $g_i \circ g_j$ is concave over $[0, +\infty)$.

Also, it is

$$(g_i(g_j(x)))'' = g_i''(g_j(x)) (g_j'(x))^2 + g_i'(g_j(x)) g_j''(x) \in C(\mathbb{R}), \quad x \in \mathbb{R}.$$

So $g_i \circ g_j$ is a sigmoid activation function with asymptotes $g_i(\pm 1)$.

Next we consider the function

$$\Phi_{i,j}(x) := \frac{1}{4} (g_i \circ g_j(x+1) - g_i \circ g_j(x-1)) > 0, \quad \forall x \in \mathbb{R}.$$

We observe that

$$\begin{aligned} \Phi_{i,j}(-x) &= \frac{1}{4} (g_i(g_j(-x+1)) - g_i(g_j(-x-1))) = \\ &= \frac{1}{4} (g_i(g_j(-(x-1))) - g_i(g_j(-(x+1)))) = \\ &= \frac{1}{4} (g_i(-g_j(x-1)) - g_i(-g_j(x+1))) = \\ &= \frac{1}{4} (-g_i(g_j(x-1)) + g_i(g_j(x+1))) = \\ &= \frac{1}{4} (g_i g_j(x+1) - g_i g_j(x-1)) = \Phi_{i,j}(x), \end{aligned}$$

that is

$$\Phi_{i,j}(-x) = \Phi_{i,j}(x), \quad \forall x \in \mathbb{R}.$$

So $\Phi_{i,j}$ can serve as a density function in general.

So we have $g_j : \mathbb{R} \rightarrow (-1, 1)$, $g_i|_{(-1,1)} : (-1, 1) \rightarrow (-1, 1)$, and the strictly increasing function $H_{i,j} := g_i|_{(-1,1)} \circ g_j : \mathbb{R} \rightarrow (-1, 1)$, with the graph of $H_{i,j}$ containing an arc of finite length, such that $H_{i,j}(0) = 0$, starting at $(-1, g_i(g_j(-1)))$ and terminating at $(1, g_i(g_j(1)))$. We call this arc also $H_{i,j}$. In particular $H_{i,j}$ is negative and convex over $(-1, 0]$, and it is positive and concave over $[0, 1)$.

So it has compact support $[-1, 1]$ and it is like a squashing function, see [4], Ch. 1, p. 8.

We will work from now on with $|H_{i,j}|$, which has as a graph a cusp joining the points $(-1, |g_i(g_j(-1))|)$, $(0, 0)$, $(1, g_i(g_j(1)))$ and with compact support, again, $[-1, 1]$. The points $(-1, |g_i(g_j(-1))|)$, $(1, g_i(g_j(1)))$ belong to the graph of $|H_{i,j}|$ and $(0, 0)$ too.

Typically $H_{i,j}$ has a steeper slope than of g_j , but it is flatter and closer to the x -axis than g_j is, e.g. $\tanh(\tanh x)$ has asymptotes ± 0.76 , while $\tanh x$ has asymptotes ± 1 , notice that $\tanh(1) = 0.76$. Clearly $H_{i,j}$ has applications in spiking neural networks.

Here we consider functions $f : \mathbb{R} \rightarrow \mathbb{R}$ to be either continuous and bounded, or uniformly continuous.

The first modulus of continuity is given by

$$\omega_1(f, \delta) := \sup_{\substack{x, y \in \mathbb{R} : \\ |x - y| \leq \delta}} |f(x) - f(y)|, \quad \delta > 0.$$

Here we have that $\omega_1(f, \delta) < +\infty$, $\delta > 0$.

In this article we study the generalized fractional pointwise and uniform convergences with rates over the real line, to the unit operator, of the 49 different "normalized cusp neural network operators",

$$\left(A_n^{(i,j)}(f)\right)(x) := \frac{\sum_{k=-n^2}^{n^2} f\left(\frac{k}{n}\right) |H_{i,j}(n^{1-\alpha}(x - \frac{k}{n}))|}{\sum_{k=-n^2}^{n^2} |H_{i,j}(n^{1-\alpha}(x - \frac{k}{n}))|}, \quad (9)$$

where $0 < \alpha < 1$ and $x \in \mathbb{R}$, $n \in \mathbb{N}$; $i, j \in \{1, \dots, 7\}$, $i \neq j$ or $i = j$.

Notice $A_n^{(i,j)}$ is a positive linear operator with $A_n^{(i,j)}(1) = 1$.

The terms in the ratio of sums (9) can be nonnegative and make sense, iff $0 < |n^{1-\alpha}(x - \frac{k}{n})| \leq 1$, i.e. $0 < |x - \frac{k}{n}| \leq \frac{1}{n^{1-\alpha}}$, iff

$$nx - n^\alpha \leq x \leq nx + n^\alpha, \quad x \neq \frac{k}{n}. \quad (10)$$

In order to have the desired order of numbers

$$-n^2 \leq nx - n^\alpha \leq nx + n^\alpha \leq n^2, \quad x \neq \frac{k}{n}, \quad (11)$$

it is sufficient to assume that

$$n \geq 1 + |x|, \quad x \neq \frac{k}{n}. \quad (12)$$

When $x \in [-1, 1]$, $x \neq \frac{k}{n}$, it is enough to assume $n \geq 2$, which implies (11), and $x \neq \frac{k}{n}$.

But the unique case $x = \frac{k}{n}$ contributes nothing and can be ignored.

Thus, without loss of generality we can take always that $x \neq \frac{k}{n}$.

Proposition 1. ([2]) Let $a, b \in \mathbb{R}$, $a \leq b$. Let $\text{card}(k) (\geq 0)$ be the maximum number of integers contained in $[a, b]$. Then

$$\max(0, (b-a) - 1) \leq \text{card}(k) \leq (b-a) + 1.$$

Note 1. We would like to establish a lower bound on $\text{card}(k)$ over the interval $[nx - n^\alpha, nx + n^\alpha]$. By Proposition 1 we get that

$$\text{card}(k) \geq \max(2n^\alpha - 1, 0).$$

We obtain $\text{card}(k) \geq 1$, if $2n^\alpha - 1 \geq 1$ iff $n \geq 1$, which is always true.

So to have the desired order (11) and $\text{card}(k) \geq 1$ over $[nx - n^\alpha, nx + n^\alpha]$, it is enough to consider

$$n \geq \max(1 + |x|, 1) = 1 + |x|. \quad (13)$$

Also notice that $\text{card}(k) \rightarrow +\infty$, as $n \rightarrow +\infty$.

Denote by $[\cdot]$ the integral part of a number and $\lceil \cdot \rceil$ its ceiling.

Thus, it is clear that

$$\left(A_n^{(i,j)}(f)\right)(x) := \frac{\sum_{k=\lceil nx-n^\alpha \rceil}^{\lceil nx+n^\alpha \rceil} f\left(\frac{k}{n}\right) |H_{i,j}(n^{1-\alpha}(x - \frac{k}{n}))|}{\sum_{k=\lceil nx-n^\alpha \rceil}^{\lceil nx+n^\alpha \rceil} |H_{i,j}(n^{1-\alpha}(x - \frac{k}{n}))|}, \quad (14)$$

$0 < \alpha < 1$, and $n \in \mathbb{N} : n \geq 1 + |x|$, $x \in \mathbb{R}$; all $i, j \in \{1, \dots, 7\}$, $i \neq j$ or $i = j$.

3 About Generalized Fractional Calculus

We will use

Definition 2.([5]) Let $\beta > 0$, $[\beta] = N$, $[\cdot]$ the ceiling of the number. Let $f \in C^N([a, b], X)$, where $[a, b] \subset \mathbb{R}$, and $(X, \|\cdot\|)$ is a Banach space. Let $g \in C^1([a, b])$, strictly increasing, such that $g^{-1} \in C^N([g(a), g(b)])$.

We define the left generalized g -fractional derivative X -valued of f of order β as follows:

$$\left(D_{*a;g}^\beta f\right)(x) := \frac{1}{\Gamma(N-\beta)} \int_a^x (g(x) - g(t))^{N-\beta-1} g'(t) (f \circ g^{-1})^{(N)}(g(t)) dt, \quad (15)$$

$\forall x \in [a, b]$. The last integral is of Bochner type.

If $\beta \notin \mathbb{N}$, by [5], we have that $\left(D_{*a;g}^\beta f\right) \in C([a, b], X)$.

We set

$$D_{*a;g}^N f(x) := \left((f \circ g^{-1})^{(N)} \circ g\right)(x) \in C([a, b], X), \quad N \in \mathbb{N}, \quad (16)$$

$$D_{*a;g}^0 f(x) = f(x), \quad \forall x \in [a, b].$$

When $g = id$, then

$$D_{*a;g}^\beta f = D_{*a;id}^\beta f = D_{*a}^\beta f, \quad (17)$$

the usual left X -valued Caputo fractional derivative, see [9, 11].

We will use

Definition 3.([5]) Let $\beta > 0$, $[\beta] = N$, $[\cdot]$ the ceiling of the number. Let $f \in C^N([a, b], X)$, where $[a, b] \subset \mathbb{R}$, and $(X, \|\cdot\|)$ is a Banach space. Let $g \in C^1([a, b])$, strictly increasing, such that $g^{-1} \in C^N([g(a), g(b)])$.

We define the right generalized g -fractional derivative X -valued of f of order β as follows:

$$\left(D_{b-;g}^\beta f\right)(x) := \frac{(-1)^N}{\Gamma(N-\beta)} \int_x^b (g(t) - g(x))^{N-\beta-1} g'(t) (f \circ g^{-1})^{(N)}(g(t)) dt, \quad (18)$$

$\forall x \in [a, b]$. The last integral is of Bochner type.

If $\beta \notin \mathbb{N}$, by [5], we have that $\left(D_{b-;g}^\beta f\right) \in C([a, b], X)$.

We set

$$D_{b-;g}^N f(x) := (-1)^N \left((f \circ g^{-1})^{(N)} \circ g\right)(x) \in C([a, b], X), \quad N \in \mathbb{N}, \quad (19)$$

$$D_{b-;g}^0 f(x) = f(x), \quad \forall x \in [a, b].$$

When $g = id$, then

$$D_{b-;g}^\beta f = D_{b-;id}^\beta f = D_{b-}^\beta f, \quad (20)$$

the usual right X -valued Caputo fractional derivative, see [3, 10, 11].

We mention the following g -left generalized X -valued Taylor's formula:

Theorem 1.([5]) Let $\beta > 0$, $N = [\beta]$ and $f \in C^N([a, b], X)$, where $[a, b] \subset \mathbb{R}$, and $(X, \|\cdot\|)$ is a Banach space. Let $g \in C^1([a, b])$, strictly increasing, such that $g^{-1} \in C^N([g(a), g(b)])$. Then

$$\begin{aligned} f(x) &= f(a) + \sum_{j_*=1}^{N-1} \frac{(g(x) - g(a))^{j_*}}{j_*!} (f \circ g^{-1})^{(j_*)}(g(a)) + \\ &\quad \frac{1}{\Gamma(\beta)} \int_a^x (g(x) - g(t))^{\beta-1} g'(t) \left(D_{*a;g}^\beta f\right)(t) dt = \\ &\quad f(a) + \sum_{j_*=1}^{N-1} \frac{(g(x) - g(a))^{j_*}}{j_*!} (f \circ g^{-1})^{(j_*)}(g(a)) + \\ &\quad \frac{1}{\Gamma(\beta)} \int_{g(a)}^{g(x)} (g(x) - z)^{\beta-1} \left(\left(D_{*a;g}^\beta f\right) \circ g^{-1}\right)(z) dz, \quad \forall x \in [a, b]. \end{aligned} \quad (21)$$

We also mention the following g -right generalized X -valued Taylor's formula:

Theorem 2. ([5]) Let $\beta > 0$, $N = \lceil \beta \rceil$ and $f \in C^N([a, b], X)$, where $[a, b] \subset \mathbb{R}$, and $(X, \|\cdot\|)$ is a Banach space. Let $g \in C^1([a, b])$, strictly increasing, such that $g^{-1} \in C^N([g(a), g(b)])$. Then

$$\begin{aligned} f(x) &= f(b) + \sum_{j_*=1}^{N-1} \frac{(g(x) - g(b))^{j_*}}{j_*!} (f \circ g^{-1})^{(j_*)}(g(b)) + \\ &\quad \frac{1}{\Gamma(\beta)} \int_x^b (g(t) - g(x))^{\beta-1} g'(t) \left(D_{b-;g}^\beta f \right)(t) dt = \\ &\quad f(b) + \sum_{j_*=1}^{N-1} \frac{(g(x) - g(b))^{j_*}}{j_*!} (f \circ g^{-1})^{(j_*)}(g(b)) + \\ &\quad \frac{1}{\Gamma(\beta)} \int_{g(x)}^{g(b)} (z - g(x))^{\beta-1} \left(\left(D_{b-;g}^\beta f \right) \circ g^{-1} \right)(z) dz, \quad \forall x \in [a, b]. \end{aligned} \quad (22)$$

Integrals in (15)-(22) are of Bochner type. For the Bochner integral excellent resources are [17] and [1, pp. 422-428].

When $X = \mathbb{R}$, then the Bochner integral collapses to the Lebesgue integral. So all the above apply also to real valued functions which we treat here.

Example 1. The following function

$$f(x) = x + \sqrt{x^2 + 1}, \quad x \in \mathbb{R},$$

has derivative $f'(x) = 1 + \frac{x}{\sqrt{x^2+1}}$ with range $(0, 2)$.

Indeed we have: $1 + \frac{x}{\sqrt{x^2+1}} > 0$, iff $\frac{x}{\sqrt{x^2+1}} > -1$, iff $x > -\sqrt{x^2+1}$, iff $-x < \sqrt{x^2+1}$. The last is true if $x \geq 0$, and if $x < 0$, then $|x| < \sqrt{x^2+1}$, iff $x^2 < x^2 + 1$, iff $0 < 1$ true.

And we have $1 + \frac{x}{\sqrt{x^2+1}} < 2$, iff $\frac{x}{\sqrt{x^2+1}} < 1$, iff $x < \sqrt{x^2+1}$, if $x < 0$ the last is true, if $x > 0$ then $x^2 < x^2 + 1$, iff $0 < 1$ true. Thus, f is strictly increasing, differentiable and uniformly continuous. It has C^1 inverse, i.e. f^{-1} is continuous and differentiable, with $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$, by continuous f' and $f' \neq 0$.

Example 2. Let $f(x) = x + \arctan(x)$. Then $f'(x) = 1 + \frac{1}{1+x^2}$, with $1 < f'(x) \leq 2$, $\forall x \in \mathbb{R}$. Thus f is C^1 , strictly increasing and uniformly continuous.

By the inverse function theorem f^{-1} is differentiable and $(f^{-1})'$ is continuous.

Example 3. Let $f(x) = 2x + \sin(x)$, it is $1 \leq f'(x) \leq 3$, thus f is uniformly continuous, strictly increasing, continuously differentiable with a C^1 inverse.

So our assumption on g in Definitions 2, 3 and Theorems 1, 2 is possible.

4 Main Results

We present the following generalized fractional approximation result.

Theorem 3. Let $\beta > 0$, $\beta \notin \mathbb{N}$, $N = \lceil \beta \rceil$. Let also a strictly increasing $g \in C^1(\mathbb{R})$, such that $g^{-1} \in C^N(\mathbb{R})$; g is also assumed to be uniformly continuous. Let also $f \in C^N(\mathbb{R})$ and assume that $(f \circ g^{-1})^{(N)} \circ g \in L_\infty(\mathbb{R})$. Let also $x \in \mathbb{R}$, $n \in \mathbb{N} : n \geq 1 + |x|$. We further assume that $D_{*x;g}^\beta f$, $D_{x-;g}^\beta f$ are uniformly continuous or continuous and bounded on $[x, +\infty)$, $(-\infty, x]$, respectively. Then

1)

$$\begin{aligned} \max_{\substack{i,j \in \{1, \dots, 7\}, \\ i \neq j \text{ or } i=j}} \left| A_n^{(i,j)}(f)(x) - f(x) - \sum_{j_*=1}^{N-1} \frac{(f \circ g^{-1})^{(j_*)}(g(x))}{j_*!} \right. \\ \left. \left(\frac{\sum_{k=\lceil nx-n^\alpha \rceil}^{\lceil nx+n^\alpha \rceil} (g(\frac{k}{n}) - g(x))^{j_*} |H_{i,j}(n^{1-\alpha}(x - \frac{k}{n}))|}{\sum_{k=\lceil nx-n^\alpha \rceil}^{\lceil nx+n^\alpha \rceil} |H_{i,j}(n^{1-\alpha}(x - \frac{k}{n}))|} \right) \right| \leq \end{aligned} \quad (23)$$

$$\frac{\omega_1^\beta \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta+1)} \left[\omega_1 \left(D_{x-;g}^\beta f; \frac{1}{n^{1-\alpha}} \right)_{(-\infty, x]} + \omega_1 \left(D_{*x;g}^\beta f; \frac{1}{n^{1-\alpha}} \right)_{[x, +\infty)} \right],$$

above it is $\sum_{j=1}^0 \cdot = 0$.

2) If $N = 1$ or $(f \circ g^{-1})^{(j)}(g(x)) = 0$, $j = 1, \dots, N-1$, we have

$$\max_{\substack{i,j \in \{1, \dots, 7\}, \\ i \neq j \text{ or } i=j}} \left| A_n^{(i,j)}(f)(x) - f(x) \right| \leq \quad (24)$$

$$\frac{\omega_1^\beta \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta+1)} \left[\omega_1 \left(D_{x-;g}^\beta f; \frac{1}{n^{1-\alpha}} \right)_{(-\infty, x]} + \omega_1 \left(D_{*x;g}^\beta f; \frac{1}{n^{1-\alpha}} \right)_{[x, +\infty)} \right],$$

3)

$$\begin{aligned} \max_{\substack{i,j \in \{1, \dots, 7\}, \\ i \neq j \text{ or } i=j}} \left| A_n^{(i,j)}(f)(x) - f(x) \right| &\leq \sum_{j_*=1}^{N-1} \frac{|(f \circ g^{-1})^{(j_*)}(g(x))|}{j_*!} \omega_1^{j_*} \left(g, \frac{1}{n^{1-\alpha}} \right) \\ &+ \frac{\omega_1^\beta \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta+1)} \left[\omega_1 \left(D_{x-;g}^\beta f; \frac{1}{n^{1-\alpha}} \right)_{(-\infty, x]} + \omega_1 \left(D_{*x;g}^\beta f; \frac{1}{n^{1-\alpha}} \right)_{[x, +\infty)} \right]. \end{aligned} \quad (25)$$

Here we get generalized fractionally with rates the pointwise convergence of $A_n^{(i,j)}(f)(x) \rightarrow f(x)$, as $n \rightarrow +\infty$, $x \in \mathbb{R}$; $i, j \in \{1, \dots, 7\}$.

Proof. Let $x \in \mathbb{R}$. We have that $D_{x-;g}^\beta f(x) = D_{*x;g}^\beta f(x) = 0$, explanation at the end of this proof.

From Theorem 6, by using the left generalized Caputo fractional Taylor formula we get that

$$\begin{aligned} f\left(\frac{k}{n}\right) &= \sum_{j_*=0}^{N-1} \frac{(f \circ g^{-1})^{(j_*)}(g(x))}{j_*!} \left(g\left(\frac{k}{n}\right) - g(x) \right)^{j_*} + \\ &\frac{1}{\Gamma(\beta)} \int_x^{\frac{k}{n}} \left(g\left(\frac{k}{n}\right) - g(J) \right)^{\beta-1} g'(J) \left[\left(D_{*x;g}^\beta f \right)(J) - \left(D_{*x;g}^\beta f \right)(x) \right] dJ, \end{aligned} \quad (26)$$

for all $x \leq \frac{k}{n} \leq x + n^{\alpha-1}$, iff $\lceil nx \rceil \leq k \leq \lfloor nx + n^\alpha \rfloor$, where $k \in \mathbb{Z}$.

Also from Theorem 7, by using the right generalized Caputo fractional Taylor formula we get

$$\begin{aligned} f\left(\frac{k}{n}\right) &= \sum_{j_*=0}^{N-1} \frac{(f \circ g^{-1})^{(j_*)}(g(x))}{j_*!} \left(g\left(\frac{k}{n}\right) - g(x) \right)^{j_*} + \\ &\frac{1}{\Gamma(\beta)} \int_{\frac{k}{n}}^x \left(g(J) - g\left(\frac{k}{n}\right) \right)^{\beta-1} g'(J) \left[\left(D_{x-;g}^\beta f \right)(J) - \left(D_{x-;g}^\beta f \right)(x) \right] dJ, \end{aligned} \quad (27)$$

for all $x - n^{\alpha-1} \leq \frac{k}{n} \leq x$, iff $\lceil nx - n^\alpha \rceil \leq k \leq \lfloor nx \rfloor$, where $k \in \mathbb{Z}$.

Notice that $\lceil nx \rceil \leq \lfloor nx \rfloor + 1$.

Call

$$W_{i,j}(x) := \sum_{k=\lceil nx-n^\alpha \rceil}^{\lfloor nx+n^\alpha \rfloor} \left| H_{i,j} \left(n^{1-\alpha} \left(x - \frac{k}{n} \right) \right) \right|, \quad i, j \in \{1, \dots, 7\}.$$

Hence we have

$$\begin{aligned} \frac{f\left(\frac{k}{n}\right) |H_{i,j}(n^{1-\alpha}(x - \frac{k}{n}))|}{W_{i,j}(x)} &= \\ \sum_{j_*=0}^{N-1} \frac{(f \circ g^{-1})^{(j_*)}(g(x))}{j_*!} \left(g\left(\frac{k}{n}\right) - g(x) \right)^{j_*} \frac{|H_{i,j}(n^{1-\alpha}(x - \frac{k}{n}))|}{W_{i,j}(x)} &+ \end{aligned} \quad (28)$$

$$\frac{|H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)\Gamma(\beta)} \int_x^{\frac{k}{n}} \left(g\left(\frac{k}{n}\right) - g(J)\right)^{\beta-1} g'(J) \left[D_{*x;g}^\beta f(J) - D_{*x;g}^\beta f(x)\right] dJ,$$

and

$$\begin{aligned} & \frac{f\left(\frac{k}{n}\right)|H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)} = \\ & \sum_{j_*=0}^{N-1} \frac{(f \circ g^{-1})^{(j_*)}(g(x))}{j_*!} \left(g\left(\frac{k}{n}\right) - g(x)\right)^{j_*} \frac{|H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)} + \\ & \frac{|H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)\Gamma(\beta)} \int_{\frac{k}{n}}^x \left(g(J) - g\left(\frac{k}{n}\right)\right)^{\beta-1} g'(J) \left[D_{x-;g}^\beta f(J) - D_{x-;g}^\beta f(x)\right] dJ. \end{aligned} \quad (29)$$

Therefore we obtain

$$\begin{aligned} & \frac{\sum_{k=[nx]+1}^{[nx+n^\alpha]} f\left(\frac{k}{n}\right)|H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)} = \\ & \sum_{j_*=0}^{N-1} \frac{(f \circ g^{-1})^{(j_*)}(g(x))}{j_*!} \left(\frac{\sum_{k=[nx]+1}^{[nx+n^\alpha]} \left(g\left(\frac{k}{n}\right) - g(x)\right)^{j_*} |H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)} \right) \\ & + \sum_{k=[nx]+1}^{[nx+n^\alpha]} \frac{|H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)\Gamma(\beta)} \\ & \int_x^{\frac{k}{n}} \left(g\left(\frac{k}{n}\right) - g(J)\right)^{\beta-1} g'(J) \left[D_{*x;g}^\beta f(J) - D_{*x;g}^\beta f(x)\right] dJ, \end{aligned} \quad (30)$$

and

$$\begin{aligned} & \frac{\sum_{k=[nx-n^\alpha]}^{[nx]} f\left(\frac{k}{n}\right)|H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)} = \\ & \sum_{j_*=0}^{N-1} \frac{(f \circ g^{-1})^{(j_*)}(g(x))}{j_*!} \left(\frac{\sum_{k=[nx-n^\alpha]}^{[nx]} \left(g\left(\frac{k}{n}\right) - g(x)\right)^{j_*} |H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)} \right) \\ & + \sum_{k=[nx-n^\alpha]}^{[nx]} \frac{|H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)\Gamma(\beta)} \\ & \int_{\frac{k}{n}}^x \left(g(J) - g\left(\frac{k}{n}\right)\right)^{\beta-1} g'(J) \left[D_{x-;g}^\beta f(J) - D_{x-;g}^\beta f(x)\right] dJ. \end{aligned} \quad (31)$$

Adding the two equalities (30) and (31) we obtain

$$\begin{aligned} & \left(A_n^{(i,j)}(f)\right)(x) = \sum_{j_*=0}^{N-1} \frac{(f \circ g^{-1})^{(j_*)}(g(x))}{j_*!} \\ & \left(\frac{\sum_{k=[nx-n^\alpha]}^{[nx+n^\alpha]} \left(g\left(\frac{k}{n}\right) - g(x)\right)^{j_*} |H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)} \right) + \widehat{M}_n^{(i,j)}(x), \end{aligned} \quad (32)$$

where $i, j \in \{1, \dots, 7\}$, and

$$\widehat{M}_n^{(i,j)}(x) := \sum_{k=[nx-n^\alpha]}^{[nx]} \frac{|H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)\Gamma(\beta)} \quad (33)$$

$$\int_{\frac{k}{n}}^x \left(g(J) - g\left(\frac{k}{n}\right) \right)^{\beta-1} g'(J) \left[D_{x-;g}^{\beta} f(J) - D_{x-;g}^{\beta} f(x) \right] dJ +$$

$$\sum_{k=[nx]+1}^{[nx+n^{\alpha}]} \frac{|H_{i,j}(n^{1-\alpha}(x - \frac{k}{n}))|}{W_{i,j}(x) \Gamma(\beta)}$$

$$\int_x^{\frac{k}{n}} \left(g\left(\frac{k}{n}\right) - g(J) \right)^{\beta-1} g'(J) \left[D_{*x;g}^{\beta} f(J) - D_{*x;g}^{\beta} f(x) \right] dJ.$$

We call

$$\widehat{M}_{1n}^{(i,j)}(x) := \sum_{k=[nx-n^{\alpha}]}^{[nx]} \frac{|H_{i,j}(n^{1-\alpha}(x - \frac{k}{n}))|}{W_{i,j}(x) \Gamma(\beta)} \quad (34)$$

$$\int_{\frac{k}{n}}^x \left(g(J) - g\left(\frac{k}{n}\right) \right)^{\beta-1} g'(J) \left[D_{x-;g}^{\beta} f(J) - D_{x-;g}^{\beta} f(x) \right] dJ,$$

and

$$\widehat{M}_{2n}^{(i,j)}(x) := \sum_{k=[nx]+1}^{[nx+n^{\alpha}]} \frac{|H_{i,j}(n^{1-\alpha}(x - \frac{k}{n}))|}{W_{i,j}(x) \Gamma(\beta)} \quad (35)$$

$$\int_x^{\frac{k}{n}} \left(g\left(\frac{k}{n}\right) - g(J) \right)^{\beta-1} g'(J) \left[D_{*x;g}^{\beta} f(J) - D_{*x;g}^{\beta} f(x) \right] dJ.$$

That is

$$\widehat{M}_n^{(i,j)}(x) = \widehat{M}_{1n}^{(i,j)}(x) + \widehat{M}_{2n}^{(i,j)}(x), \quad (36)$$

where $i, j \in \{1, \dots, 7\}$.

Furthermore we get

$$E_n^{(i,j)}(f)(x) := \left(A_n^{(i,j)}(f) \right)(x) - f(x) - \sum_{j_*=1}^{N-1} \frac{(f \circ g^{-1})^{(j_*)}(g(x))}{j_*!}$$

$$\left(\frac{\sum_{k=[nx-n^{\alpha}]}^{[nx+n^{\alpha}]} \left(g\left(\frac{k}{n}\right) - g(x) \right)^{j_*} |H_{i,j}(n^{1-\alpha}(x - \frac{k}{n}))|}{W_{i,j}(x)} \right) = \widehat{M}_n^{(i,j)}(x), \quad (37)$$

that is

$$\left| E_n^{(i,j)}(f)(x) \right| \leq \left| \widehat{M}_n^{(i,j)}(x) \right| \leq \left| \widehat{M}_{1n}^{(i,j)}(x) \right| + \left| \widehat{M}_{2n}^{(i,j)}(x) \right|. \quad (38)$$

If $N = 1$, then

$$E_n^{(i,j)}(f)(x) = A_n^{(i,j)}(f)(x) - f(x) = \widehat{M}_n^{(i,j)}(x). \quad (39)$$

If $(f \circ g^{-1})^{(j_*)}(g(x)) = 0$, for $j_* = 1, \dots, N-1$, then again

$$E_n^{(i,j)}(f)(x) = A_n^{(i,j)}(f)(x) - f(x) = \widehat{M}_n^{(i,j)}(x), \quad (40)$$

where $i, j \in \{1, \dots, 7\}$.

We can also have

$$\left| A_n^{(i,j)}(f)(x) - f(x) \right| \leq \sum_{j_*=1}^{N-1} \frac{|(f \circ g^{-1})^{(j_*)}(g(x))|}{j_*!} \omega_1^{j_*} \left(g, \frac{1}{n^{1-\alpha}} \right) + \widehat{M}_n^{(i,j)}(x), \quad (41)$$

by g being also uniformly continuous we have that $\omega_1 \left(g, \frac{1}{n^{1-\alpha}} \right) < +\infty$.

We observe that

$$\begin{aligned}
 \left| \widehat{M}_{1n}^{(i,j)}(x) \right| &\leq \sum_{k=\lceil nx-n^\alpha \rceil}^{\lfloor nx \rfloor} \frac{|H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)\Gamma(\beta)} \\
 &\int_{\frac{k}{n}}^x \left(g(J) - g\left(\frac{k}{n}\right) \right)^{\beta-1} g'(J) \left| D_{x-;g}^\beta f(J) - D_{x-;g}^\beta f(x) \right| dJ \leq \\
 &\sum_{k=\lceil nx-n^\alpha \rceil}^{\lfloor nx \rfloor} \frac{|H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)\Gamma(\beta)} \\
 &\int_{\frac{k}{n}}^x \left(g(J) - g\left(\frac{k}{n}\right) \right)^{\beta-1} g'(J) \omega_1 \left(D_{x-;g}^\beta f, |J-x| \right)_{(-\infty, x]} dJ \leq \\
 &\sum_{k=\lceil nx-n^\alpha \rceil}^{\lfloor nx \rfloor} \frac{|H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)\Gamma(\beta)} \\
 &\left(\int_{g(\frac{k}{n})}^{g(x)} \left(g(J) - g\left(\frac{k}{n}\right) \right)^{\beta-1} dg(J) \right) \omega_1 \left(D_{x-;g}^\beta f, \left| x - \frac{k}{n} \right| \right)_{(-\infty, x]} \leq \\
 &\left(\sum_{k=\lceil nx-n^\alpha \rceil}^{\lfloor nx \rfloor} \frac{|H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)\Gamma(\beta)} \frac{(g(x) - g(\frac{k}{n}))^\beta}{\beta} \right) \omega_1 \left(D_{x-;g}^\beta f, \frac{1}{n^{1-\alpha}} \right)_{(-\infty, x]} \quad (42)
 \end{aligned}$$

(by assuming g is uniformly continuous, then $\omega_1(g, \delta) < +\infty, \forall \delta > 0$)

$$\begin{aligned}
 &\leq \left(\sum_{k=\lceil nx-n^\alpha \rceil}^{\lfloor nx \rfloor} \frac{|H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)} \right) \frac{\omega_1^\beta \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta+1)} \omega_1 \left(D_{x-;g}^\beta f, \frac{1}{n^{1-\alpha}} \right)_{(-\infty, x]} \\
 &\leq \left(\frac{\sum_{k=\lceil nx-n^\alpha \rceil}^{\lfloor nx \rfloor} |H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)} \right) \frac{\omega_1^\beta \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta+1)} \omega_1 \left(D_{x-;g}^\beta f, \frac{1}{n^{1-\alpha}} \right)_{(-\infty, x]} \quad (44) \\
 &= \frac{\omega_1^\beta \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta+1)} \omega_1 \left(D_{x-;g}^\beta f, \frac{1}{n^{1-\alpha}} \right)_{(-\infty, x]}.
 \end{aligned}$$

Thus it holds

$$\left| \widehat{M}_{1n}^{(i,j)}(x) \right| \leq \frac{\omega_1^\beta \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta+1)} \omega_1 \left(D_{x-;g}^\beta f, \frac{1}{n^{1-\alpha}} \right)_{(-\infty, x]}. \quad (45)$$

Next we estimate

$$\begin{aligned}
 \left| \widehat{M}_{2n}^{(i,j)}(x) \right| &\leq \frac{\sum_{k=\lfloor nx \rfloor+1}^{\lfloor nx+n^\alpha \rfloor} |H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)\Gamma(\beta)} \\
 &\int_x^{\frac{k}{n}} \left(g\left(\frac{k}{n}\right) - g(J) \right)^{\beta-1} g'(J) \left| D_{*x;g}^\beta f(J) - D_{*x;g}^\beta f(x) \right| dJ \leq \\
 &\frac{\sum_{k=\lfloor nx \rfloor+1}^{\lfloor nx+n^\alpha \rfloor} |H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)\Gamma(\beta)} \\
 &\int_x^{\frac{k}{n}} \left(g\left(\frac{k}{n}\right) - g(J) \right)^{\beta-1} g'(J) \omega_1 \left(D_{*x;g}^\beta f, |J-x| \right)_{[x, +\infty)} dJ \leq \\
 &\frac{\sum_{k=\lfloor nx \rfloor+1}^{\lfloor nx+n^\alpha \rfloor} |H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)\Gamma(\beta)} \quad (46)
 \end{aligned}$$

(47)

$$\begin{aligned} & \left(\int_{g(x)}^{g(\frac{k}{n})} \left(g\left(\frac{k}{n}\right) - g(J) \right)^{\beta-1} dg(J) \right) \omega_1 \left(D_{*x;g}^{\beta} f, \left| \frac{k}{n} - x \right| \right)_{[x,+\infty)} \leq \\ & \left(\frac{\sum_{k=\lfloor nx \rfloor+1}^{\lfloor nx+n^{\alpha} \rfloor} |H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x) \Gamma(\beta)} \frac{(g(\frac{k}{n}) - g(x))^{\beta}}{\beta} \right) \omega_1 \left(D_{*x;g}^{\beta} f, \frac{1}{n^{1-\alpha}} \right)_{[x,+\infty)} \leq \end{aligned} \quad (48)$$

$$\begin{aligned} & \left(\frac{\sum_{k=\lfloor nx \rfloor+1}^{\lfloor nx+n^{\alpha} \rfloor} |H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)} \right) \frac{\omega_1^{\beta} \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta+1)} \omega_1 \left(D_{*x;g}^{\beta} f, \frac{1}{n^{1-\alpha}} \right)_{[x,+\infty)} \leq \\ & \left(\frac{\sum_{k=\lfloor nx-n^{\alpha} \rfloor}^{\lfloor nx+n^{\alpha} \rfloor} |H_{i,j}(n^{1-\alpha}(x-\frac{k}{n}))|}{W_{i,j}(x)} \right) \frac{\omega_1^{\beta} \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta+1)} \omega_1 \left(D_{*x;g}^{\beta} f, \frac{1}{n^{1-\alpha}} \right)_{[x,+\infty)} \quad (49) \\ & = \frac{\omega_1^{\beta} \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta+1)} \omega_1 \left(D_{*x;g}^{\beta} f, \frac{1}{n^{1-\alpha}} \right)_{[x,+\infty)}. \end{aligned}$$

Thus

$$\left| \widehat{M}_{2n}^{(i,j)}(x) \right| \leq \frac{\omega_1^{\beta} \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta+1)} \omega_1 \left(D_{*x;g}^{\beta} f, \frac{1}{n^{1-\alpha}} \right)_{[x,+\infty)}. \quad (50)$$

We have found that

$$\begin{aligned} & \left| \widehat{M}_n^{(i,j)}(x) \right| \leq \frac{\omega_1^{\beta} \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta+1)} \\ & \left[\omega_1 \left(D_{x-;g}^{\beta} f, \frac{1}{n^{1-\alpha}} \right)_{(-\infty,x]} + \omega_1 \left(D_{*x;g}^{\beta} f, \frac{1}{n^{1-\alpha}} \right)_{[x,+\infty)} \right] < +\infty, \end{aligned} \quad (51)$$

where $i, j \in \{1, \dots, 7\}$.

Here we assumed that $(f \circ g^{-1})^{(N)} \circ g \in L_{\infty}(\mathbb{R})$.

Next we get

$$\begin{aligned} \left| D_{*x;g}^{\beta} f(s) \right| & \leq \frac{1}{\Gamma(N-\beta)} \int_x^s (g(s) - g(t))^{N-\beta-1} g'(t) \left| (f \circ g^{-1})^{(N)}(g(t)) \right| dt \leq \\ & \frac{\left\| (f \circ g^{-1})^{(N)} \circ g \right\|_{\infty, \mathbb{R}}}{\Gamma(N-\beta)} \int_{g(x)}^{g(s)} (g(s) - g(t))^{N-\beta-1} dg(t) = \\ & \frac{\left\| (f \circ g^{-1})^{(N)} \circ g \right\|_{\infty, \mathbb{R}}}{\Gamma(N-\beta+1)} (g(s) - g(x))^{N-\beta}. \end{aligned} \quad (52)$$

That is

$$\left| D_{*x;g}^{\beta} f(s) \right| \leq \frac{\left\| (f \circ g^{-1})^{(N)} \circ g \right\|_{\infty, \mathbb{R}}}{\Gamma(N-\beta+1)} (g(s) - g(x))^{N-\beta}, \quad (53)$$

$\forall s \in [x, +\infty)$.

Clearly then

$$D_{*x;g}^{\beta} f(x) = 0. \quad (54)$$

Acting similarly we have:

$$\begin{aligned} \left| D_{x-;g}^{\beta} f(s) \right| & \leq \frac{1}{\Gamma(N-\beta)} \int_s^x (g(t) - g(s))^{N-\beta-1} g'(t) \left| (f \circ g^{-1})^{(N)}(g(t)) \right| dt \leq \\ & \frac{\left\| (f \circ g^{-1})^{(N)} \circ g \right\|_{\infty, \mathbb{R}}}{\Gamma(N-\beta)} \int_{g(s)}^{g(x)} (g(t) - g(s))^{N-\beta-1} dg(t) = \end{aligned} \quad (55)$$

$$\frac{\left\| (f \circ g^{-1})^{(N)} \circ g \right\|_{\infty, \mathbb{R}}}{\Gamma(N - \beta + 1)} (g(x) - g(s))^{N - \beta}.$$

That is

$$\left| D_{x-;g}^{\beta} f(s) \right| \leq \frac{\left\| (f \circ g^{-1})^{(N)} \circ g \right\|_{\infty, \mathbb{R}}}{\Gamma(N - \beta + 1)} (g(x) - g(s))^{N - \beta}, \quad (56)$$

$\forall s \in (-\infty, x]$.

Clearly then

$$D_{x-;g}^{\beta} f(x) = 0. \quad (57)$$

As an application of Theorem 3 we get:

Theorem 4. All as in Theorem 3, and under the assumption that $D_{*x;g}^{\beta} f(t)$, $D_{x-;g}^{\beta} f(t)$ are both continuous and bounded in $(x, t) \in \mathbb{R}^2$; $n \in \mathbb{N} : n \geq 2$. Then

1) if $N = 1$ we have

$$\max_{\substack{i, j \in \{1, \dots, 7\}, \\ i \neq j \text{ or } i = j}} \left\| A_n^{(i, j)}(f) - f \right\|_{\infty, [-1, 1]} \leq \frac{\omega_1^{\beta} \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta + 1)} \left[\sup_{x \in [-1, 1]} \omega_1 \left(D_{x-;g}^{\beta} f; \frac{1}{n^{1-\alpha}} \right)_{(-\infty, x]} + \sup_{x \in [-1, 1]} \omega_1 \left(D_{*x;g}^{\beta} f; \frac{1}{n^{1-\alpha}} \right)_{[x, +\infty)} \right], \quad (58)$$

2) in general we have

$$\max_{\substack{i, j \in \{1, \dots, 7\}, \\ i \neq j \text{ or } i = j}} \left\| A_n^{(i, j)}(f) - f \right\|_{\infty, [-1, 1]} \leq \sum_{j_*=1}^{N-1} \frac{\left\| (f \circ g^{-1})^{(j_*)} \circ g \right\|_{\infty, [-1, 1]}}{j_*!} \omega_1^{j_*} \left(g, \frac{1}{n^{1-\alpha}} \right) + \frac{\omega_1^{\beta} \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta + 1)} \left[\sup_{x \in [-1, 1]} \omega_1 \left(D_{x-;g}^{\beta} f; \frac{1}{n^{1-\alpha}} \right)_{(-\infty, x]} + \sup_{x \in [-1, 1]} \omega_1 \left(D_{*x;g}^{\beta} f; \frac{1}{n^{1-\alpha}} \right)_{[x, +\infty)} \right]. \quad (59)$$

An interesting case is when $\beta = \frac{1}{2}$.

Here we get generalized fractionally with rates the uniform convergence of $A_n^{(i, j)}(f) \rightarrow f$, over $[-1, 1]$, as $n \rightarrow +\infty$; $i, j \in \{1, \dots, 7\}$.

Proof. By Theorem 3, and by

$$\left\| D_{*x;g}^{\beta} f \right\|_{\infty} \leq K_1, \quad \forall x \in \mathbb{R};$$

and

$$\left\| D_{x-;g}^{\beta} f \right\|_{\infty} \leq K_2, \quad \forall x \in \mathbb{R},$$

where $K_1, K_2 > 0$.

We get that

$$\omega_1 \left(D_{x-;g}^{\beta} f; \xi \right)_{(-\infty, x]} \leq 2K_2,$$

and

$$\omega_1 \left(D_{*x;g}^{\beta} f; \xi \right)_{[x, +\infty)} \leq 2K_1, \quad \forall x \in \mathbb{R},$$

for any $\xi \geq 0$.

It follows

Corollary 1. Let $\beta > 0$, $\beta \notin \mathbb{N}$, $N = \lceil \beta \rceil$. Let also a strictly increasing $g \in C^1(\mathbb{R})$, such that $g^{-1} \in C^N(\mathbb{R})$; g is also assumed to be uniformly continuous. Let also $f \in C^N(\mathbb{R})$ and assume that $(f \circ g^{-1})^{(N)} \circ g \in L_\infty(\mathbb{R})$. Let $x \in [-\phi, \phi]$, $\phi > 0$ and $n \in \mathbb{N} : n \geq 1 + \phi$, $0 < \alpha < 1$. Here $D_{*x;g}^\beta f(t)$, $D_{x-;g}^\beta f(t)$ are both continuous and bounded in $(x, t) \in \mathbb{R}^2$. Then

1) If $N = 1$, we have

$$\max_{\substack{i,j \in \{1, \dots, 7\}, \\ i \neq j \text{ or } i=j}} \left| A_n^{(i,j)}(f)(x) - f(x) \right| \leq \frac{\omega_1^\beta \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta + 1)} \quad (60)$$

$$\left[\sup_{x \in [-\phi, \phi]} \omega_1 \left(D_{x-;g}^\beta f; \frac{1}{n^{1-\alpha}} \right)_{(-\infty, x]} + \sup_{x \in [-\phi, \phi]} \omega_1 \left(D_{*x;g}^\beta f; \frac{1}{n^{1-\alpha}} \right)_{[x, +\infty)} \right],$$

2) in general we have

$$\begin{aligned} \max_{\substack{i,j \in \{1, \dots, 7\}, \\ i \neq j \text{ or } i=j}} \left| A_n^{(i,j)}(f)(x) - f(x) \right| &\leq \sum_{j_*=1}^{N-1} \frac{\left| (f \circ g^{-1})^{(j_*)}(g(x)) \right|}{j_*!} \omega_1^{j_*} \left(g, \frac{1}{n^{1-\alpha}} \right) \\ &+ \frac{\omega_1^\beta \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta + 1)} \left[\sup_{x \in [-\phi, \phi]} \omega_1 \left(D_{x-;g}^\beta f; \frac{1}{n^{1-\alpha}} \right)_{(-\infty, x]} + \sup_{x \in [-\phi, \phi]} \omega_1 \left(D_{*x;g}^\beta f; \frac{1}{n^{1-\alpha}} \right)_{[x, +\infty)} \right]. \end{aligned} \quad (61)$$

Proof. By Theorem 3.

By Corollary 1, and monotonicity and subadditivity properties of $\|\cdot\|_p$, $p \geq 1$, we obtain the following L_p result.

Corollary 2. Here all as in Corollary 1 and $p \geq 1$. Then

1) If $N = 1$, we have

$$\max_{\substack{i,j \in \{1, \dots, 7\}, \\ i \neq j \text{ or } i=j}} \left\| A_n^{(i,j)} f - f \right\|_{p, [-\phi, \phi]} \leq 2^{\frac{1}{p}} \phi^{\frac{1}{p}} \frac{\omega_1^\beta \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta + 1)} \quad (62)$$

$$\left[\sup_{x \in [-\phi, \phi]} \omega_1 \left(D_{x-;g}^\beta f; \frac{1}{n^{1-\alpha}} \right)_{(-\infty, x]} + \sup_{x \in [-\phi, \phi]} \omega_1 \left(D_{*x;g}^\beta f; \frac{1}{n^{1-\alpha}} \right)_{[x, +\infty)} \right],$$

and

2) in general we get

$$\begin{aligned} \max_{\substack{i,j \in \{1, \dots, 7\}, \\ i \neq j \text{ or } i=j}} \left\| A_n^{(i,j)} f - f \right\|_{p, [-\phi, \phi]} &\leq \sum_{j_*=1}^{N-1} \frac{\left\| (f \circ g^{-1})^{(j_*)} \circ g \right\|_{p, [-\phi, \phi]}}{j_*!} \omega_1^{j_*} \left(g, \frac{1}{n^{1-\alpha}} \right) \\ &+ 2^{\frac{1}{p}} \phi^{\frac{1}{p}} \frac{\omega_1^\beta \left(g, \frac{1}{n^{1-\alpha}} \right)}{\Gamma(\beta + 1)} \end{aligned} \quad (63)$$

$$\left[\sup_{x \in [-\phi, \phi]} \omega_1 \left(D_{x-;g}^\beta f; \frac{1}{n^{1-\alpha}} \right)_{(-\infty, x]} + \sup_{x \in [-\phi, \phi]} \omega_1 \left(D_{*x;g}^\beta f; \frac{1}{n^{1-\alpha}} \right)_{[x, +\infty)} \right].$$

By (62), (63) we derive the generalized fractional L_p , $p \geq 1$, convergence with rates of $A^{(i,j)} f$ to f , all $i, j \in \{1, \dots, 7\}$.

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